

25WS67 Coursework 2: Verification and Validation of a Remotely Operated Electric Car Ferry
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Abstract

The Woolwich Ferry has taken delivery of two diesel-electric ferries that have been beset with reliability issues. While they are now in service, they continue to be unavailable, with single boat operation being common. This presents an opportunity to provide an alternative vehicle to service the route. As a result, an initial architecture and requirements set for a battery-electric remotely operated unmanned vehicle (ROUV) has been developed. To ensure reliability and performance, along with meeting stakeholder expectations, this work lays out a verification and validation implementation using SysML1.5. Requirements have been validated, a plan for verification has been drafted, coverage has been examined, and preliminary verification has been performed to identify non-compliance early and to help explore the design space as it progresses.

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1 Introduction

1.1 Background

The Woolwich Ferry is a free ferry service operated by Transport for London which connects Woolwich, on the southern bank of the river Thames, and North Woolwich, on the northern bank of the Thames[1]. It carries passengers and road vehicles across the Thames at intervals of 15 minutes, taking 5 minutes to make the journey[2]. In 2018, the service took delivery of two new diesel-electric ferries. These vessels have been beset with issues, and the Mayor of London has issued an apology for their condition[3]. As a result, there is the possibility of replacement vessels being required in the near future.

Following previous work[4], this opportunity could be targeted by a remotely operated unmanned vehicle (ROUV) with autonomous capability to reduce crew strain in tedious tasks such as mooring and unmooring. Beyond that, moving to a battery electric propulsion system would allow for the service to have significantly reduced carbon emissions and for operating costs to remain stable despite significant fluctuations in petroleum-based fuel prices.

To do this, a ferry must be designed and this design must then undergo a verification and validation (V & V) programme. This means that a number of models will need to be created to understand the effect of different design variables on the performance of the vessel and its compliance against its requirements.

When designing a vessel, a key consideration is that of the power needed to maintain a given speed through water. Power is defined as the work done over a given time, which can then be shown to be a function of constant force and velocity:

$$P = \frac{dE}{dt} = \frac{d}{dt} (\vec{F} \cdot s) \Rightarrow P = F \frac{ds}{dt} \\ \Rightarrow P = Fu \quad (1)$$

Where P is power, E is energy (or work done), t is time, $\vec{F}$ is the resultant force on the vessel, s is the distance and u is the velocity of the vessel. When considering the forces on the vessel:

$$\vec{F} = \vec{F}_p - \vec{F}_d$$

Where $\vec{F}_p$ is the propulsive force, and $\vec{F}_d$ is the drag force. Newton's well-known second law states that:

$$\vec{F} = m\vec{a} \Rightarrow \vec{a} = \frac{\vec{F}}{m}$$

Where m is the non-zero mass of an object and $\vec{a}$ its acceleration. For a constant speed:

$$\vec{a} \equiv 0 \Rightarrow \vec{F} = 0 \Rightarrow \vec{F}_p - \vec{F}_d = 0 \\ \Rightarrow \vec{F}_p = \vec{F}_d \quad (2)$$

Knowing that the the required force for a given speed is equal to the drag force, that drag force must be calculated. The force of drag can be found[5] as:

$$F_d = \frac{1}{2} \rho u^2 A_s C_D \Rightarrow C_D = \frac{F_d}{\frac{1}{2} \rho u^2 A_s} \quad (3)$$

Where ρ is the density of the fluid (in this case, water), A_s is the wetted surface area, and C_D the non-dimensionalised drag coefficient. C_D is composed of three constituent parts, but at Froud numbers ($Fr = \frac{uL}{g}$, L is the length of the vessel, g is acceleration due to gravity) less than 0.25:

$$C_D = C_V$$

Where C_V is the coefficient of viscous resistance. This is a reasonable assumption for the ROUV, and serves to simplify the models greatly. C_V can be further decomposed[6][7] as $C_V = C_F + KC_F$, where C_F is the coefficient of friction and K is the 'hull form factor.' Finally, C_F and K can be calculated as follows:

$$C_F = \frac{0.075}{((\log_{10} Re) - 2)^2} \quad (4)$$

$$K \approx 19 \left(\frac{\nabla}{LBT} \cdot \frac{B}{L} \right)^2 \quad (5)$$

Where ∇ is the volume of displacement. B the breadth of the hull, T the draught, and $Re = \frac{Lu}{\nu}$ the non-dimensional Reynolds number of the flow.

With $A_S \approx 1.025 \left(\frac{\nabla}{T} + 1.7LT \right)$ [5], the drag force and thus power can be found.

1.2 Problem Statement

As mentioned above, there is an opportunity for a ferry with novel propulsion and autonomous capability to be entered into service as the Woolwich Ferry. Doing so would require a physical architecture, a requirements set (both functional and non-functional), and a V & V test plan to ensure compliance and that the ROUV would meet stakeholder expectations. The current class of ferry is unreliable and expensive, and service is frequently affected. Despite the diesel-hybrid propulsion, fluctuations in the price of oil may have an effect on the operating costs of the ferry, and as oil reserves run dry these ferries are not well suited for a long service life. Any solution will need to be carefully validated against stakeholder requirements, and verification of the system requirements will help avoid another class of ships that are unreliable and costly.

1.3 System Narrative

The ROUV design shall have a battery electric powertrain, and be largely similar in performance to the ships that preceded the current generation[8]. This ROUV shall have a requirements set of both functional and non-functional requirements to inform the design of a physical architecture. The system shall be validated against the stakeholder needs in a manner that is traceable throughout. A focus will be given to performance, to ensure continuous running of the service, and to reliability.

1.4 Aims and Objectives

This report aims to provide an initial plan for the verification of the system, with validation of requirements and appropriate models informing testing and further design.

To do this, the report has the following objects:

1. To generate an initial requirements set and physical architecture for verification and validation and subsequent design.
2. To ascertain stakeholder needs and validate the requirements subset against them, considering both customers, operators, and passengers.
3. To produce test procedures for a subset of important requirements and relate them to both requirements but also physical architecture.
4. To plan out the verification methods and tests that will verify the requirements, along with performing initial analysis on those requirements to see where design variables can be altered.
5. To provide an indication as to the failure modes of the system, their importance, and ways of mitigating them.

2 Technical Results

2.1 Requirements Validation

To begin the verification and validation process, the requirements defined alongside the architecture were validated. Specifically, they were validated against the stakeholder needs. A set of high level needs were defined, and were related to the requirements where fit. The mapping of the needs to the requirements can be seen in figure 1. As seen, all of the stakeholder needs are satisfied by at least one requirement. Additionally, all requirements are satisfying a stakeholder need, or contain a requirement that does this (the hierarchy of the requirements is not visible in this view). There is however a notable exception in that the "Remote Control" stakeholder need is entirely unsatisfied. This is because the ship was designed primarily as a "dumb" vessel for simplicity, which could then have remote control capability retrofitted to it. Despite this, to enable remote control a much higher bandwidth communications link must be provided than that mandated by Requirement ID 7.1.

Source	Target	Requirements:AIS	Requirements:Anchor Control	Requirements:Astern Acceleration	Requirements:Availability	Requirements:Cargo	Requirements:Charging	Requirements:Charging Speed	Requirements:Communications	Requirements:Depth Measurement	Requirements:Dimension Envelope	Requirements:Emergency Stop	Requirements:Endurance	Requirements:Extreme Astern Deceleration	Requirements:Extreme Endurance	Requirements:Extreme Forward Acceleration	Requirements:Extreme Temperature	Requirements:Forward Acceleration	Requirements:Heading Control	Requirements:Heading Control at 0 kts	Requirements:Lateral Movement	Requirements:Longitudinal Movement	Requirements:Maximum Draught	Requirements:Navigation	Requirements:Normal Temperature	Requirements:Passenger Capacity	Requirements:Passenger Transport	Requirements:Power	Requirements:Prime Mover Control	Requirements:Proximity Feedback	Requirements:Radar Tracking	Requirements:Rudder Control	Requirements:Safety	Requirements:Situational Awareness	Requirements:Structure	Requirements:Thruster Control	Requirements:Top Speed Through Water	Requirements:Vehicle Capacity	Requirements:Vehicle Transport	Requirements:VHF	Requirements:Visual Feedback		
Stakeholder Needs:Cargo Capacity																																											
Stakeholder Needs:Communication																																											
Stakeholder Needs:Manoeuvrability																																											
Stakeholder Needs:Navigation																																											
Stakeholder Needs:Remote Control																																											
Stakeholder Needs:Safety																																											
Stakeholder Needs:Service Availability																																											
Stakeholder Needs:Service Performance																																											
Stakeholder Needs:Stability																																											

Figure 1: Requirement Validation - Stakeholder Needs

Additionally, the requirements were validated against existing data. For example, the maximum permissible draught of 1.2 metres (Requirement ID 4.1) was validated against charts provided by the Port of London Authority[9]. It shows that the minimum depth along the ferry's route is at worst 2.1 metres, meaning that 1.2 metres is sufficiently shallow to avoid grounding even at low tide. This was further corroborated by inspection of draught lines on the sides of the previous and current generations of the Woolwich ferry showing a draught of between 1.2 and 1.4 metres[10]. This inspection also verified the dimension envelope requirement (Requirement ID 4.2), where Google Maps was used to estimate the dimensions of the current generation of ferry and was used to inform the dimension envelope[11]. A final example is the value of Top Speed Through Water (Requirement ID 1.1.1.3), which is the maximum permissible speed in the region that the ferry operates (between Wandsworth Bridge and Margaretness Point) without special dispensation as set out in [12].

Validating the requirements against stakeholder needs in this way provides traceability all through the model, relating the requirements, test cases and contexts, and if further developed the design back to the stakeholder. This facilitates the defence of any changes and design decisions that arise as a result of design or the V & V process. However, these stakeholder needs are predictive and therefore unlikely to be comprehensive, and iterative engagement with the stakeholders to understand their worldviews and needs would benefit them greatly and lead to a more justified requirements model.

2.2 Test Case Definitions

Following the validation, a number of test cases were developed. At this initial level, the requirements deemed most important had their test cases defined. As this is a method of public transport, those requirements which relate directly to scheduling of the ferry have been selected. These are the forward and astern accelerations (Requirement IDs 1.1.1.1, 1.1.1.2 and children), the top speed through water (Requirement ID 1.1.1.3), the endurance (Requirement ID 2.1.1 and 2.1.1.1), and the charging speed (Requirement ID 2.2.1).

To do this, three test contexts were defined: one for speed and acceleration, one for endurance, and one for charging speed (see figure 2). Ideally, these would be stereotyped as «systemContext», which is a

non-normative extension of «block». This was not implemented in Enterprise Architect 15.1, and defining this stereotype is without the scope of this work. As such, «block» was chosen to stereotype the test contexts. These contexts represent the environment that the test case is a part of, and to manage complexity it has been used to capture the operations, constraints, parts, and values that are relevant to the test cases.

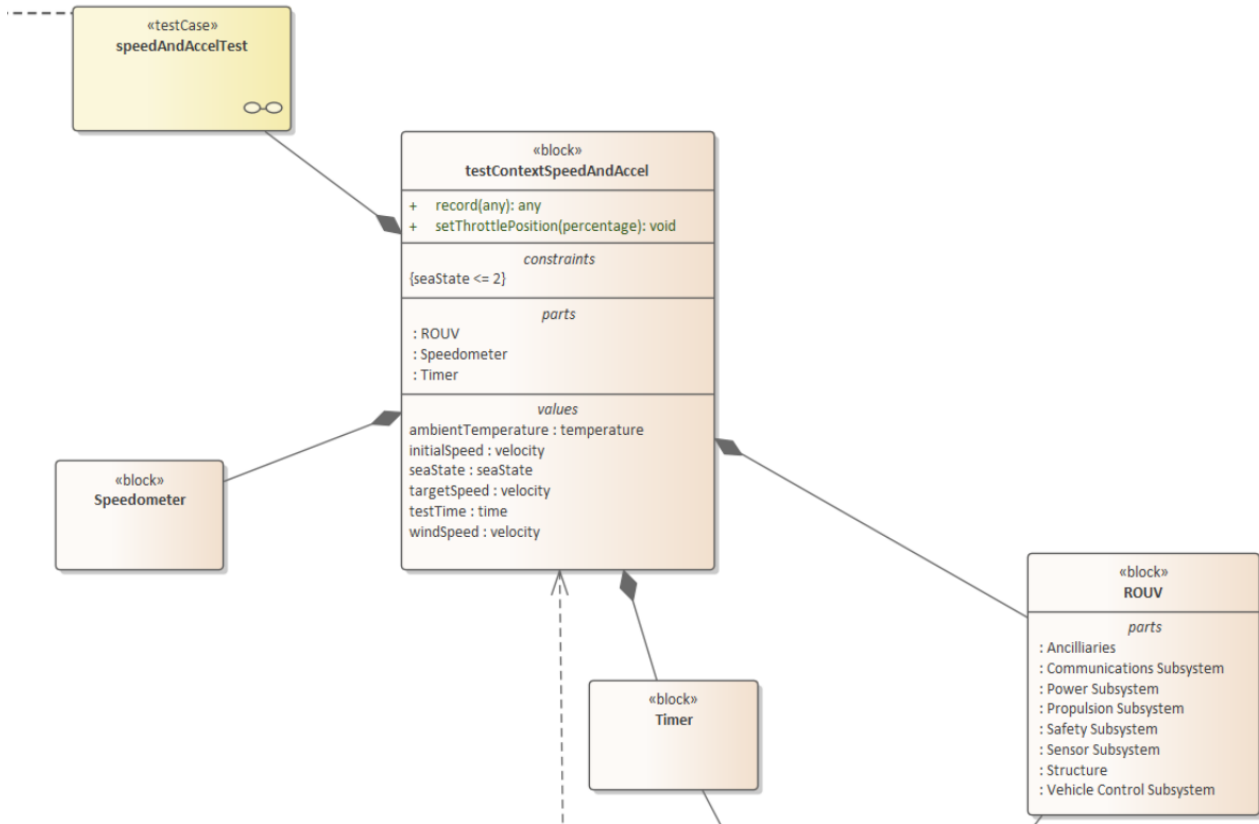


Figure 2: Example Test Context - Speed and Acceleration

The test cases are each activities that read test contexts and other parameters, and then return the verdict. The speedAndAccelTest is based off BS ISO 13643-4:2017[13], which defines how to test a ship's ability to accelerate among other manoeuvres. This has been extended to determine top speed by re-applying full throttle to accelerate to maximum speed through water. The standard mandates that the speed be held for 2 minutes (modelled by a timer event element) and only passes if the variation is less than 2% over that time. This prevents either an error in the measuring equipment or a surge from swell approaching from the stern of the vessel from falsely resulting in a pass.

Aligning the above test with the standard lends credence to it and gives confidence in the results. Unfortunately, no representative standard was found that would accurately determine the endurance or the charging speed of the vessel. This is likely because battery-powered ships may still a niche offering, and because the standards that do exist focus on the lifetime of the battery instead of the single-run range, which is without the scope of this work. As a result, a simulation test has been developed to determine the endurance. This was not possible for the above, as hydrodynamic forces are dominant in governing the acceleration and speed (see section 2.4). In this case however, to manage cost and to allow testing before integration and production of an entire vessel, a single drive motor is run with a representative resistive load placed in series with it and the battery. This test takes a defined speed profile, which defines the power delivered to the motor for a corresponding speed through water (found analytically). This profile time, along with the charge remaining and the known nominal capacity, are fed into an operation that calculates the maximum endurance, which in turn is compared to the acceptable value and determines the output. Choosing this approach allows for a wide variety of representative loads and profiles to be input, enabling simply comparison of design variables when searching for optimum values.

Finally, the charging test simply reads the battery from the system context and charges in, monitoring the current state of charge until it exceeds 80%. This is then recorded and compared to the acceptable charging time.

This test also implements a pre-condition, ensuring the battery's state of charge is 20% and removing the need to describe this in the procedure.

A common feature of these tests is the use of four verdict kinds. The pass and fail kinds are mandatory, but the inclusion of both inconclusive and error allow for the test to monitor itself and ensures that the result is representative and useful. Additionally, all tests record their initial conditions read from the test context for this reason, or for further analysis to see if these ambient conditions can effect the outcomes.

These test cases provide good coverage over the requirements most important for scheduling of the ferry and for the number of vessels needed to maintain constant service. If the system was to be developed further, use of the International Towing Tank Conference's (ITTC) test procedures[7] on scale models of the vessel should be used to define further test cases that can allow verification of the manoeuvring requirements before sea trial.

2.3 Verification Compliance Matrix

With test procedures defined, a Verification Compliance Matrix (VCM) was drafted. This sets out the testing plan, and was used to inform which requirements could be tested in this work (2.4) So that traceability throughout the model from method to requirement is visible, blocks representing different types of testing were defined and related to the requirements using the «satisfy» relationship. Doing so also allows for the model elements to automatically construct the VCM, which then shows coverage of requirements and reflects any changes made to the model. The VCM can be seen in figure 3

Source	Target																																											
	Requirements: AIS	Requirements: Anchor Control	Requirements: Astern Acceleration	Requirements: Availability	Requirements: Cargo	Requirements: Churning	Requirements: Churning Speed	Requirements: Communications	Requirements: Depth Measurement	Requirements: Dimension Envelope	Requirements: Emergency Stop	Requirements: Endurance	Requirements: Extreme Astern Deceleration	Requirements: Extreme Endurance	Requirements: Extreme Forward Acceleration	Requirements: Extreme Temperature	Requirements: Forward Acceleration	Requirements: Heading Control	Requirements: Heading Control at 0 Its	Requirements: Lateral Movement	Requirements: Longitudinal Movement	Requirements: Maximum Draught	Requirements: Navigation	Requirements: Normal Temperature	Requirements: Passenger Capacity	Requirements: Passenger Transport	Requirements: Power	Requirements: Prime Mover Control	Requirements: Proximity Feedback	Requirements: Radar Tracking	Requirements: Rudder Control	Requirements: Safety	Requirements: Situational Awareness	Requirements: Structure	Requirements: Thruster Control	Requirements: Top Speed Through Water	Requirements: Vehicle Capacity	Requirements: Vehicle Transport	Requirements: VHF	Requirements: Visual Feedback				
Verification Methods: Analysis			f	f	f	f	f			f			f	f	f	f	f						f		f	f																		
Verification Methods: Design Review	f	f				f		f	f	f	f							f	f	f	f	f				f	f	f	f	f	f													
Verification Methods: Literature Review	f	f	f	f	f	f	f	f	f	f	f	f	f	f	f	f	f	f	f	f	f	f	f	f	f	f	f	f	f	f	f	f	f	f	f	f	f	f	f	f	f	f	f	f
Verification Methods: Model/Tow Tank			f							f	f							f	f		f	f																						
Verification Methods: Sea Demonstration			f		f				f	f	f	f	f					f	f	f	f		f	f	f	f	f	f	f	f	f	f	f											
Verification Methods: Sea Trial	f	f	f	f	f	f	f	f	f	f	f	f	f	f	f	f	f	f	f	f	f	f	f	f	f	f	f	f	f	f	f	f	f	f	f	f	f	f	f	f	f	f	f	f
Verification Methods: Test Rig				f		f	f						f			f																												

Figure 3: Verification Compliance Matrix

From the VCM we can see that literature review is applicable to all of the requirements. This makes sense, as existing literature informs the models that the other verification methods rely on. As a result, it is sensible to perform literature review of the requirements and their domains before commencing any of the other methods. It is also a relatively cost-effective method of verification, requiring no specialist equipment and low risk. Additionally, reviewing designs is appropriate for requirements that specify design variables, as if they are not properly reflected in the design there is no point proceeding to more intensive methods of verification.

No literature is fully representative of the ROUV, however, so confidence gained through review is low. Similarly, the design simply stating that it is compliant is not sufficient. Thus, it is necessary to generate models of the system and use their outputs as verification. To maintain low risk at this stage, analysis is the next method that should be attempted. As above, the literature review provides models for use in both hand calculations and software verification, which is the focus of sections 2.4 and 2.5. As the VCM shows, however, analysis is not applicable to all of the requirements. Analysis does not often verify functional requirements, and in some cases defining models is not feasible, such as in verifying communication characteristics. Moreover, even though analysis provides more confidence than reviews, the models may not be representative of reality in all cases.

To gain more confidence, models and test rigs can be used to see the effects of certain environmental conditions on performance in reality. The model makes the distinction between model/towing tank testing, and test rig testing. This is because towing tank specifically tests the dynamics of the ROUV, whilst the test rig tests elements that can operate with simple models of other subsystems. The cost of these methods is significant compared to any of the preceding methods, requiring specialist facilities and expertise to run the majority of these tests. There is also an increase in risk, with components potentially failing and needing replacement.

The greatest level of confidence is drawn from sea trials, which tests every subsystem post-integration in the expected operating environment. As can be seen from the VCM, every requirement is tested in sea trials, as there is a risk that the interaction between subsystems aboard the vessel may lead to emergent behaviour that none of the previous methods will have identified. The cost and risk here is highest, with the need to build the vessel, have it certified, and operated along with the potential loss of the platform. It is a mandatory part of the V & V process, however, as it is the final chance for non-compliances to be identified. The cost and risk can also be offset by separating the tests into alongside tests, which are cheaper and safer, and at-sea tests.

Finally, the sea demonstration shows key functionality and performance to the customer. A selection of requirements are thus verified with this method, as exhaustive testing would be time-consuming and likely unwanted from the stakeholder's point of view. This is the highest-risk method, as failure would result in embarrassment in front of stakeholders and would put into question the results produced by the rest of the process. As such, doing it after exhaustive sea-trials is vital.

2.4 Demonstration Testing

Demonstration testing in this work has consisted of software analysis following literature review. This testing was to map an initial set of design variables to compliances. Testing and analysis was undertaken using Python 3.14, using the `numpy`, `scipy`, `pandas`, and `matplotlib` modules. A `utilities.py` module was written, containing helper functions that were used by both the `demonstrationTest.py` script and the `determinationTest.py` script (see section 2.5) The script runs through testing top speed, acceleration, deceleration, endurance, and charging time, when varying ship hull geometry, cruising speed, battery sizes, and ambient temperatures. Its results are printed to STDOUT which can then be recorded in a plaintext file using bash shell redirection. The results of the testing can be seen in figure 4.

```

1  ===ROUV Demonstration Test===
2  Re = 246933331.2, Fr = 0.2544976408726206
3
4  Power surplus = -361820.3020257009 Watts.
5  Requirement ID 1.1.1.3 Pass: False
6
7  Time to cruise: 3.7114817190784404 Seconds.
8  Requirement ID 1.1.1.1 Pass: True
9
10 Time to stop: 8.548777107548563 Seconds.
11 Requirement ID 1.1.1.2 Pass: True
12
13 Endurance: 1.9021476233627512 hours.
14 Requirement ID 2.1.1 Pass: True
15
16 1.4899381844187902
17 Extreme Endurance: 1.4899381844187902 Hours.
18 Requirement ID 2.1.1.1 Pass: True
19
20 Charging Time: 1.2
21 Requirement ID 2.2.1 Pass: False

```

Figure 4: Results of demonstration tests

The initial set of design variables are 'best guesses' based on the non-functional requirements (section 2.1). A variable that is not derived from there is k , defined as the Hull Profile Control Point. The model is capable of testing different hull profiles, which result in different volumes of displacement and wetted area which in turn affect the power needed to overcome drag. Each hull is defined as a symmetrical, flat bottomed hull for stability and increased displacement compared to prismatic hulls of the same draught. The sides of the hull, however, are

defined by a cubic bézier curve, whose second control point is free to move up and down the y axis between 0 (the keel) and T (the waterline). The cubic bézier curve[14] is defined as:

$$x = (1 - t)^3 x_1 + 3t(1 - t)^2 x_2 + 3t^2(1 - t)x_3 + t^3 x_4, \forall 0 \leq t \leq 1 \quad (6)$$

$$y = (1 - t)^3 y_1 + 3t(1 - t)^2 y_2 + 3t^2(1 - t)y_3 + t^3 y_4, \forall 0 \leq t \leq 1 \quad (7)$$

Where (x_i, y_i) are the coordinates of the control points, (x, y) are the coordinates of the point on the curve at t . These arrays of points are then combined into a spline, which forms the hull profile. Here, k corresponds to y_2 , which has the effect seen in figure 5. The parameterisation of the ship's hull in this way allows for flexibility in it's design, and if further analysis determines that a different hull type (e.g. prismatic) is more appropriate, the control points can be adjusted to reflect this change throughout the test. From this profile, the area under the line is subtracted from the area $L \cdot B \cdot T$ before being multiplied by L to get the volume of displacement[5] before being passed to a function that determines the power of the drag on the vessel. This is not entirely accurate, as the shape of the bow has been neglected for simplicity. Further work should model the bow and re-validate these requirements.

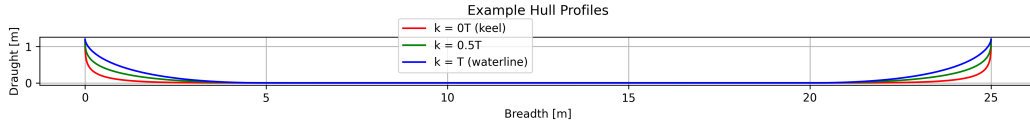


Figure 5: Representative hull profiles

After this, the acceleration requirements are verified. Each of the forward and astern accelerations are similar, with different initial and final velocities. This is calculated with a helper function that finds the power required for a sample of velocities between the initial and final, and then finds the resultant acceleration for each of these powers. It then finds the time required as (found through dimensional analysis):

$$t = \int \frac{1}{a(u)} du$$

The model assumes that the motor power is at its maximum constantly, which is reasonable for verification. In service, worse performance may be seen if the coxswain does not provide full motor power. The model is also capable of determining if a velocity is not reachable, and will raise an exception if this is the case.

Next the endurance requirement is tested. Using the above two functions, a simple power profile is defined where the vessel uses full power to accelerate, then maintains its cruising speed, before using full power to decelerate. Numerically integrating along this profile gives the energy used in a single run. Finding the number of runs from a nominal battery capacity allows for the total endurance of the vessel to be found. This is repeated

for a lower ambient temperature. To model this, a battery efficiency model was made by fitting a curve to data found in [15].

Finally, the charging time simply uses the defined charging power and required capacity to find the time.

The power profile is not ideal, as changes in power are instantaneous, which is not realistic. Additionally, the battery efficiency model is reliant on 10 samples. For a more accurate model, physical testing with a representative battery would be required.

When looking at the results, it is clear that the vessel is capable of accelerating and decelerating rapidly, and the vessel also complies with the endurance requirements. The ROUV is unable to maintain a maximum speed of 12 knots, however, requiring an extra 361 kW, and with the charging power defined the charging time is too long.

2.5 Determination Testing

To explore the design space further, determination testing was undertaken. In this, a set of results was obtained from ranges of design variables. First, the max speed requirement was investigated. Ranges of top speeds and of vessel profiles were defined, and the power required then plotted as a surface (figure 6)

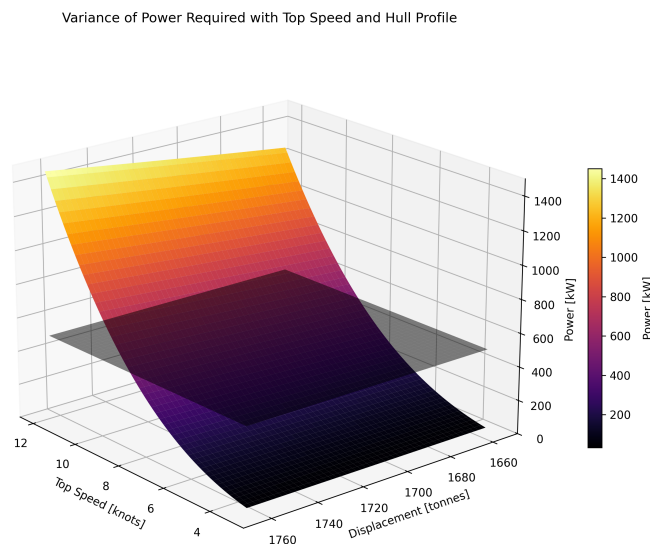


Figure 6: Response surface of Power against cruise speed and hull profile

The surf shows that the power required is strongly correlated with the cruising speed selected, but only weakly with the hull profile. From this, development of the design would see a hull chosen for largest displacement to increase cargo capacity. To comply with the top speed requirement, the requirement would need to be relaxed such that a top speed of less than 8.5 knots is required.

Additionally, sensitivities of the different design variables on endurance were identified. For each of the variables, ranges were defined and the plots of resultant endurance were generated (figures 7, 8, 9, 10). From these plots we can draw some insights: First is that the hull profile has a much smaller impact on the range than the other variables, which is consistent with the trend seen in figure 6. The trend between speed and endurance is also similar to that in figure 6. Ambient temperature has more of an impact than hull profile does, but even at the extreme (-10 degrees), the requirement would still be met. Finally, the effect of battery capacity on both battery mass and endurance has been shown. Increasing battery capacity has a significant positive impact on the range of the ROUV, but the added mass needs to be considered. The graph shows that the mass trend has a larger gradient than the endurance, so any gains in endurance would be eroded by this increasing mass. Additionally, as the mass increases, there is a risk that the amount of cargo that the ferry can carry will be reduced.

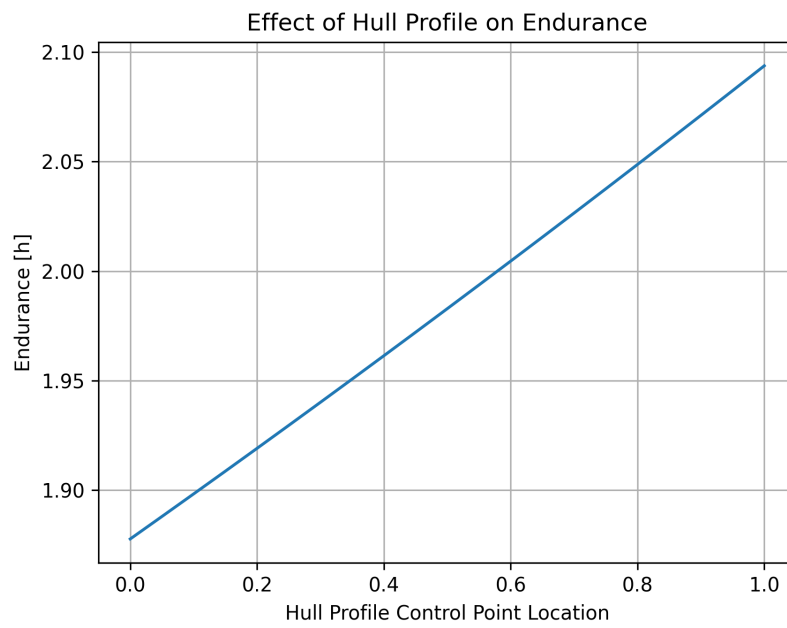


Figure 7: Hull profile sensitivity

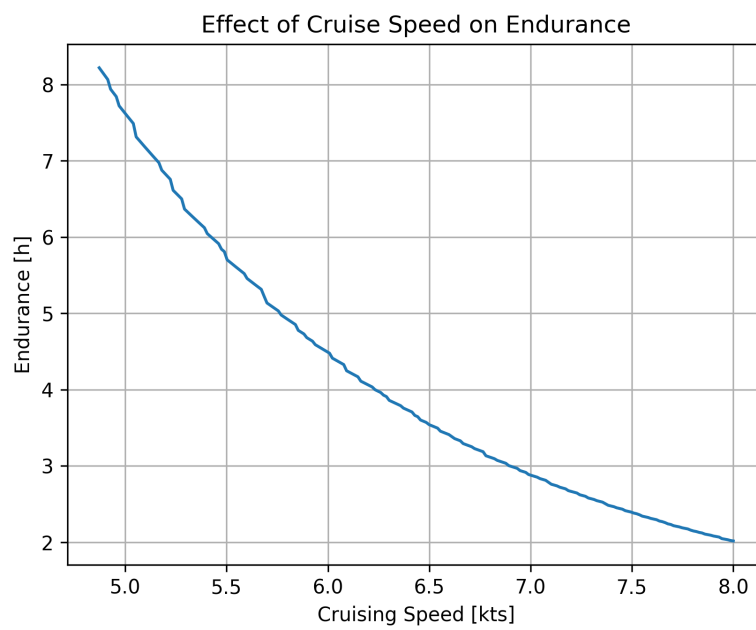


Figure 8: Cruising speed sensitivity

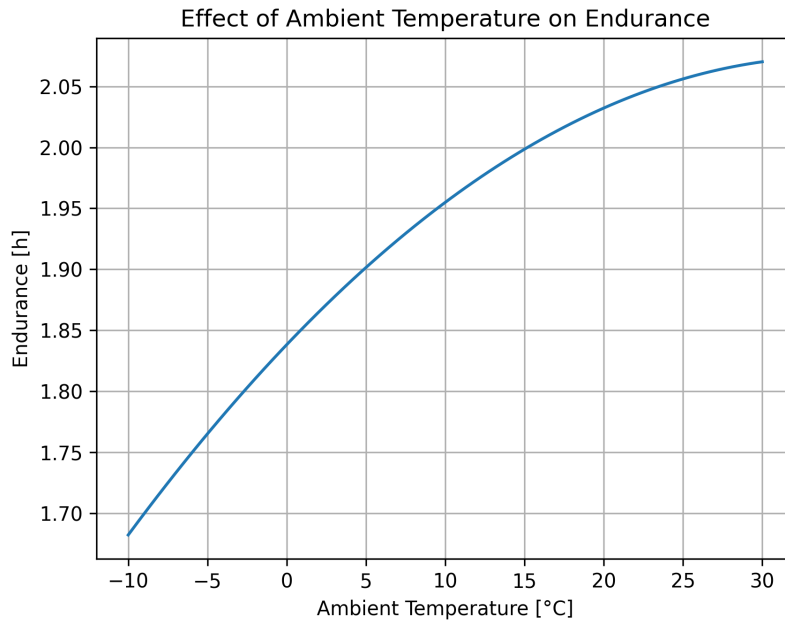


Figure 9: Temperature sensitivity

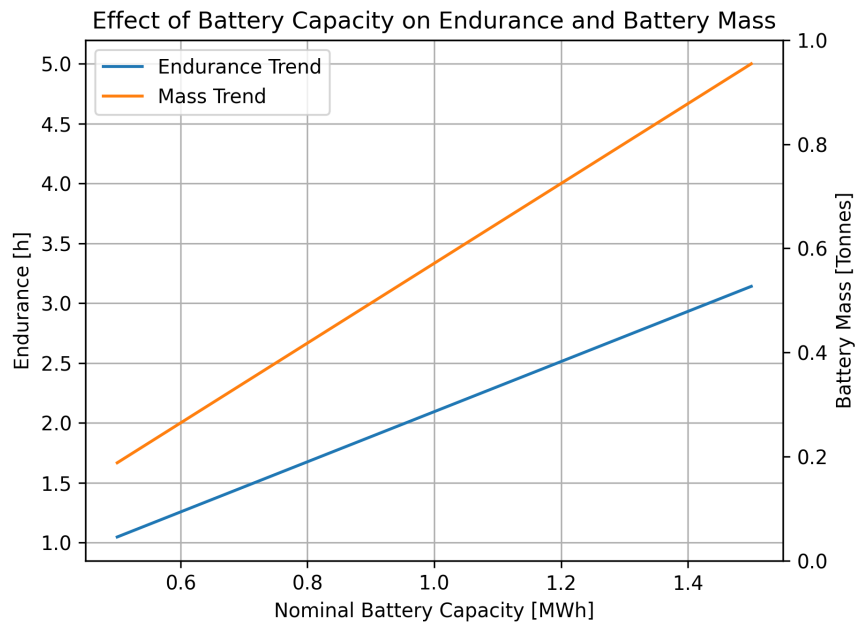


Figure 10: Battery Capacity, Endurance, and Mass sensitivity

2.6 Failure Analysis

At this stage, it is important to consider the failures that could occur with the current architecture. To do this, a failure modes and events analysis (FMEA) matrix has been generated. In this, each subsystem has had potential failure modes identified. Each one has been given a score for severity, likelihood, and preventability. These were multiplied together to produce a Risk Priority Number, and were ranked. Severities were ranked out of 10 instead of 5 so that the RPN is weighted towards the more severe failure modes. The FMEA matrix can be seen in figure 11. As the ROUV develops, revisiting the FMEA matrix and identifying failure modes at a subsystem level will provide mitigations that can be incorporated into the design iteratively.

System/Component	Failure Mode	Possible Causes	Effects on Vessel	Severity (S)	Likelihood (L)	Preventability (P)	RPN	Mitigation/Controls
Battery	Total battery failure	Thermal runaway; internal short	Loss of propulsion; blackout	10	3	4	120	Redundant battery banks; fire suppression; thermal monitoring
Battery	Battery overheating	Cooling failure; high load	Fire risk; reduced lifespan	9	4	3	108	Active cooling; temperature alarms; load limiting
Battery	State-of-charge miscalculation	BMS sensor error	Unexpected power loss mid-crossing	8	3	5	120	Redundant SOC estimation; conservative margins
Drive Motor	One motor failure	Bearing failure; winding fault	Reduced manoeuvrability; slower speed	7	4	3	84	Dual motor redundancy; condition monitoring
Drive Motor	Both motors fail	Power supply loss; control failure	Vessel adrift	10	2	4	80	Emergency propulsion procedures; tug assistance
Power Subsystem	Bus failure	Short circuit; insulation failure	Loss of propulsion and auxiliaries	10	2	5	100	Segregated bus design; protection relays
Motor Controller	Inverter failure	Component overheating; electronics fault	Loss of one or both motors	8	3	4	96	Redundant inverters; cooling systems
Vehicle Control Subsystem	Loss of control signals	Software bug; network failure	Loss of propulsion control	9	3	5	135	Redundant control systems; manual override
Vehicle Control Subsystem	Sensor failure (speed/draught)	Calibration drift; damage	Incorrect navigation decisions	6	4	6	144	Sensor redundancy; periodic calibration
Structure	Excess draught (> design)	Overloading >200 t; water ingress	Reduced clearance; grounding risk	9	3	3	81	Load monitoring; draught sensors; alarms
Structure	Hull breach	Collision; grounding	Flooding; potential sinking	10	2	3	60	Watertight compartments; bilge pumps
Stability & Loading	Uneven vehicle loading	Poor deck management	List; reduced stability	8	5	4	160	Load procedures; crew training; deck markings
Stability & Loading	Overcapacity loading	Operational error	Increased draught; structural stress	9	3	3	81	Weight checks; access control
Drive Motor	Insufficient thrust	Fouling; motor degradation	Delays; inability to manoeuvre	6	4	4	96	Maintenance; hull cleaning
Structure	Steering failure	Actuator fault; system issue	Loss of manoeuvrability	9	3	4	108	Redundant steering; emergency mode
Safety Subsystem	Collision	Human error; poor visibility	Structural damage; injury	10	3	3	90	Radar; AIS; crew training
Safety Subsystem	Battery fire	Thermal runaway	Severe damage; evacuation	10	2	3	60	Fire suppression; isolation zones
Safety Subsystem	Electrical fire	Cable fault; overload	System damage; smoke hazard	9	3	4	108	Circuit protection; inspections
Safety Subsystem	Bilge pump failure	Mechanical/electrical fault	Flooding risk	9	3	5	135	Redundant pumps; alarms

Figure 11: FMEA matrix

3 Conclusion

3.1 V&V Specification

- An initial V & V system has been modelled to support the design of a battery electric ferry.
- Three test procedures have been defined that relate to three test contexts.
- These test contexts relate to parts of the physical architecture.
- A requirement hierarchy captures the functional and non-functional requirements, along with relating them to the verification methods and stakeholder needs.
- An extensive software model performs V & V on requirements deemed most important, and enables exploration of the design space for future optimisation.
- Traceability has been maintained throughout the work, leveraging tools that model based systems engineering provides.

3.2 Achievement of Aims and Objectives

This work has achieved its objectives:

1. An initial requirements set and physical architecture has been generated.
2. Stakeholder needs have been identified and modelled, being related to the relevant requirements.
3. Three test procedures have been defined, using tests contexts to parameterise the tests and enable testing across different design variable ranges.
4. The verification plan has been defined, being informed by a VCM
5. An initial FMEA analysis has been performed, identifying failure modes and their mitigations

3.3 Highlights and Recommendations

Of particular note in this work are the software models generated, which give a high degree of confidence in the results. Additionally, the test procedure definition is rigorous and when standards were available have been informed by them to build further trust in them. It is recommended that analysis and software verification on the entire requirements set and design space be undertaken next, including a thorough literature review. Following that, discussion with stakeholders around requirements that need relaxing or modifying would result in a more compliant system whilst keeping costs minimal.

Bibliography

Of particular note are [5], [6], and [7] for a more in-depth study of hydrodynamics and shipbuilding, which may inform further verification.

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